Morning Paper Report - Friday, October 23, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Modeling Working Memory in Neurodegeneration: A Focus on EEG Methods

Y. Komarova, Alexander Zakharov, M. Sergeeva, N. Romanchuk, Tatyana Vladimirova, I. Shirolapov
Diagnostics, 2025

Source: Semantic Scholar; matched terms: working memory, modeling, neural network, artificial intelligence

Link: https://www.semanticscholar.org/paper/c268c38eee8ca52e091bcee88e0a974ede63b3c5

Goal: Working memory is one of the most vulnerable cognitive domains in neurodegenerative diseases

Method: This review aims to systematize current approaches to modeling working memory phenotypes using electroencephalography (EEG)

Results: Limitations of existing methods, especially EEG variability, are considered The review concludes by outlining future directions: integration of multimodal EEG data, application of artificial intelligence, and development of digital cognitive biomarkers for hybrid models capable of predicting cognitive decline and advancing clinical translation.

Conclusion: The review concludes by outlining future directions: integration of multimodal EEG data, application of artificial intelligence, and development of digital cognitive biomarkers for hybrid models capable of predicting cognitive decline and advancing clinical translation.

2. Set-Size Scaling Effects in Visual Working Memory Models Identify the Decoding Rule Used in Retrieval

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory

Link: https://osf.io/preprints/psyarxiv/6wk73_v1/

Goal: We examined how recent visual working memory (VWM) models for continuous-outcome tasks quantify memory capacity limitations through model parameters, and whether these quantifications conform to the inverse square-root scaling predictions derived from the sample-size model

Method: We examined how recent visual working memory (VWM) models for continuous-outcome tasks quantify memory capacity limitations through model parameters, and whether these quantifications conform to the inverse square-root scaling predictions derived from the sample-size model

Results: These results demonstrate that the apparent locus of capacity limitation in continuous VWM depends systematically on how population-level neural activity is decoded in memory retrieval and provide evidence that the form of decoding used in making decisions is identifiable from the parameter invariance in the VWM model.

Conclusion: In contrast, under maximum-output, or MAX, decoding, the signal amplitude parameter γ scaled with decreasing population size These results demonstrate that the apparent locus of capacity limitation in continuous VWM depends systematically on how population-level neural activity is decoded in memory retrieval and provide evidence that the form of decoding used in making decisions is identifiable from the parameter...

3. The Nature and Nurture of Visual Search: Genetic Predispositions, Experiential Shaping, and the Catalyst of Artificial Intelligence

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual search, deep learning, artificial intelligence

Link: https://osf.io/preprints/psyarxiv/updys_v1/

Goal: Draft version, [30/04/26]

Method: Overall, the literature suggests that visual search is fundamentally built by 'nature' (which establishes biological constraints) and optimised by 'nurture', with deep learning models providing a novel framework to compare the indispensable contributions of both.

Results: Furthermore, this paper utilises recent advancements in artificial intelligence as digital proxies to test these perspectives Overall, the literature suggests that visual search is fundamentally built by 'nature' (which establishes biological constraints) and optimised by 'nurture', with deep learning models providing a novel framework to compare the indispensable contributions of both.

Conclusion: Overall, the literature suggests that visual search is fundamentally built by 'nature' (which establishes biological constraints) and optimised by 'nurture', with deep learning models providing a novel framework to compare the indispensable contributions of both.

4. Similarity-Based Compression in Working Memory: Implications for Decay and Refreshing Models

Benjamin Kowialiewski, Benoît Lemaire, Sophie Portrat
Computational Brain & Behavior, 2024

Source: Crossref; matched terms: working memory, computational model, modeling

Link: https://doi.org/10.1007/s42113-023-00179-0

Goal: Abstract The ability to compress information is a fundamental cognitive function

Method: In addition, the model captures fine-grained patterns of transposition errors recently reported

Results: Recent evidence suggests that the similarity between items can be used to compress information, leading to a rich pattern of behavioral results

Conclusion: Recent evidence suggests that the similarity between items can be used to compress information, leading to a rich pattern of behavioral results

5. Flexible Working Memory Model in Spiking Neural Network With Two Types of Plasticity

Natalia S. Kovaleva, Valery V. Matrosov, Mikhail Mishchenko
2023

Source: Crossref; matched terms: working memory, flexibility, neural network

Link: https://doi.org/10.20944/preprints202301.0502.v1

Goal: Working memory (WM) is a brain system for short-term storage and manipulation of information and plays an important role in complex cognitive tasks

Method: We propose a WM model considering two types of plasticity: short-term plasticity and STDP

Results: In this paper, we show that such populations can be formed due to the mechanisms of spike-timing-dependent plasticity (STDP) – the phase dependence associated with the ratio of the pulse times of the interacting neurons

Conclusion: We have shown formation of neuronal clusters encoding items in the WM model, that can be formed by external stimulation of a group of neurons due to the mechanisms of STDP and hold and reactivated by short-term plasticity mechanisms The dynamic formation of neuronal clusters instead of pre-formed clusters gives additional

flexibility to the model.